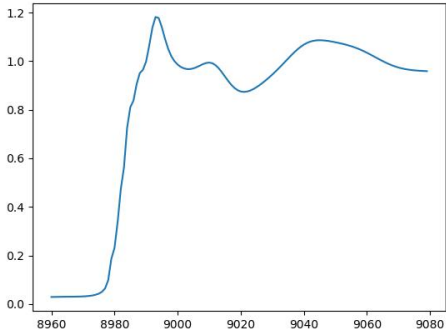


Further information of the code

1.Dataset Format Specification

Each file contains a list, where each element of the list corresponds to one 3D structure and its associated spectrum. See the table below for detailed description.

File	
z.npy	Atomic type dimensions: [number of atoms] Example dimension: 44
pos.npy	Atomic coordinates dimensions: [number of atoms, 3] Example dimensions: 44, 3
y.npy	Spectrum data mu dimensions: 120 The x-coordinate is <code>np.linspace(edge - 19, edge + 100, 120)</code>, where edge is the K-edge/L-edge of the corresponding element. Example: 

2. Directory structure

"run_one_element" :Simple example of Ni single element model.

"run_Ni50FeCoCuZn" :Data Experiment to Verify the Enhancement of XAS Predictive Capability for Ni-Deficient Samples by FeCoCuZn-Containing Samples.

"run_energyrange" :Prexanesmodel using two XAS3D model for two pre-edge feature and XANES prediction.

"run_MOE":MOE multi-expert model using XAS3D as module.

"expalin_3d4dLa":Model interpretability.

"run_fit_MnJT": XANES fitting example of Mn doped Co₃O₄ material.

3. Running Code

Copy dataset files to folder:

The URLs for the datasets and models are located at <https://nadc.china-vo.org/res/r101826/>.

Copy Ni element dataset to "run_MOE" "run_energyrange", "run_one_element", folder, or change the dataset file location in the code.

Ni element dataset:

Ni_CCDC_train_pos.npy, Ni_CCDC_train_y.npy, Ni_CCDC_train_z.npy

Ni_CCDC_valid_pos.npy, Ni_CCDC_valid_y.npy, Ni_CCDC_valid_z.npy

Ni_CCDC_test_pos.npy, Ni_CCDC_test_y.npy, Ni_CCDC_test_z.npy

Copy below dataset file for data experiment to "run_Ni50FeCoCuZn" folder.

Ni50FeCoCuZn8.5k_train_pos.npy, Ni50FeCoCuZn8.5k_train_y.npy, Ni50FeCoCuZn8.5k_train_z.npy

FeCoCuZn1k_valid_pos.npy, FeCoCuZn1k_valid_y.npy, FeCoCuZn1k_valid_z.npy

Copy atomic number file La4d3d_unconv_valid_z.npy to folder "run_expalin_3d4dLa".

Copy model file model_example_MnJT_best256_3_0.0005_5_2_82_5.pt to folder "run_fit_MnJT".

In "run_one_element" folder run:

```
python valid_XAS3Dabs_Ni_3d.py 99
```

In "run_Ni50FeCoCuZn" folder run:

```
python valid_XAS3Dabs_Ni_3d.py 99
```

In "run_energyrange" folder run:

```
python valid_prexanes.py 99
```

In "run_MOE" folder run:

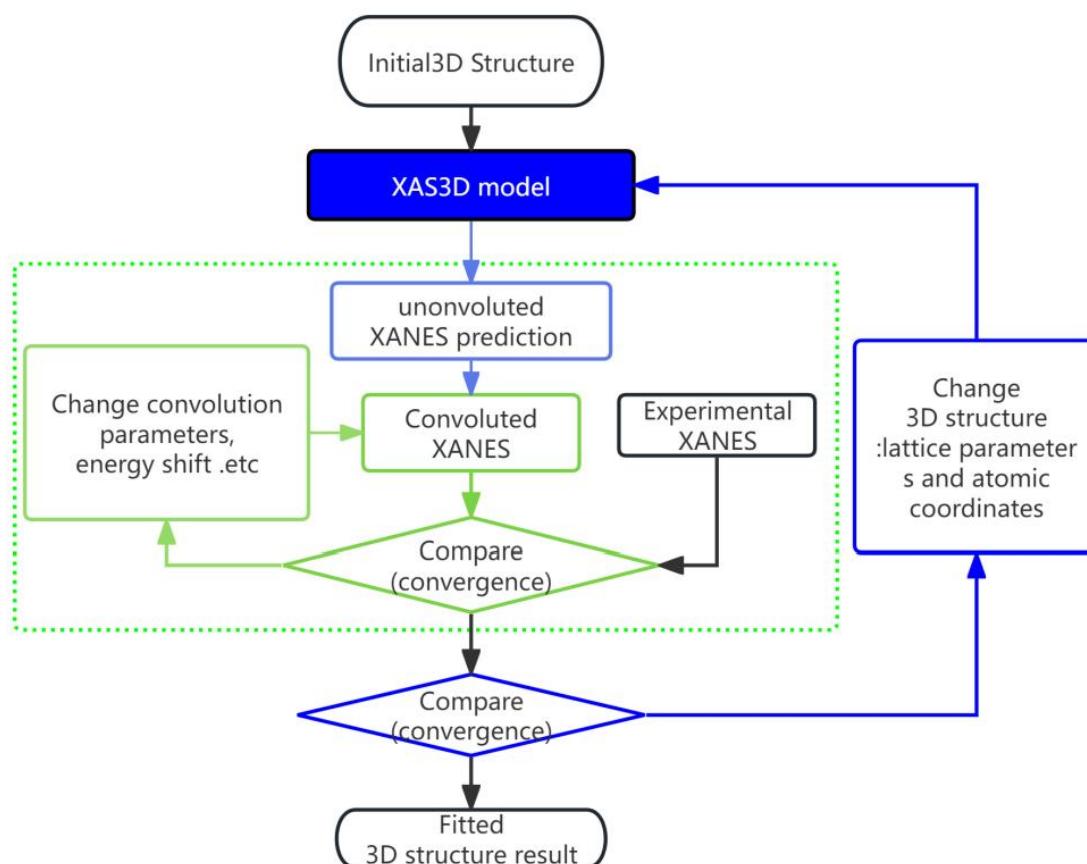
python valid_moe.py 99

In "run_fit_MnJT" folder run:

python Fit_F1_slurm_ax_plan_other.py 99

4.Fitting workflow diagram.

Here, we propose a feasible technical approach based on XANES fitting using the XAS3D model. In the modified XANES fitting workflow diagram, the XAS3D model replaces multiple scattering calculation to predict the spectrum for any proposed 3D structure. By iteratively varying the three-dimensional structure and comparing the predicted spectrum from the XAS3D model with the experimental spectrum, a structure consistent with the experimental data can be determined, thereby achieving the quantitative determination of the material's 3D structure. This fitting process comprises two nested layers: the outer layer corresponds to the aforementioned structural optimization loop, while the inner layer fits non-structural parameters such as broadening, energy shift, and normalization factor. Within the inner layer, the XAS3D model predicts the unconvoluted XANES for a given structure. This unconvoluted spectrum is then compared against the experimental spectrum after applying variable broadening, energy shifts, and amplitude scaling to determine the optimal non-structural parameters.



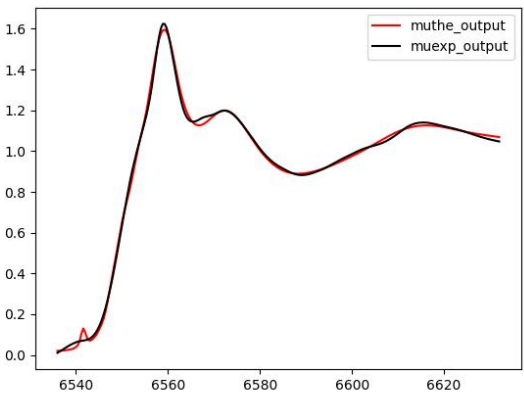
5. Fitting introduction and Mn doped material example

Fitting example of Mn doped Co₃O₄.

Copy model file model_example_MnJT_best256_3_0.0005_5_2_82_5.pt to folder

"run_fit_MnJT".

In "run_fit_MnJT" folder run "python Fit_F1_slurm_ax_plan_other.py 99"

Input file	Information
Experimental spectrum file	The first column is energy in eV, the second column is XANES. Example:JT.exp
Cartesian coordinate file	Cartesian coordinate file of the initial 3D structure, with units in Angstrom. Each line of the file contains the x, y, z coordinates of one atom, for a total of three columns. Example:Mn_start.cart
Atom type file	The atom type corresponding to each atom. Example:Mn_start.ele
Output file	Information
opt_res_e_exp_the_output.png	Plot of fitting results for quick viewing. Example: 
opt_res_e_exp_the_output.txt	Experimental spectrum and the model-predicted spectrum (broadened) corresponding to the 3D structure obtained from fitting. The first column is energy, the second column is the experimental spectrum, and the third column is the convoluted model-predicted spectrum.
opt_res_mu_pred.txt	The unconvoluted XANES predicted by the XAS3D model corresponding to the 3D results obtained from fitting.
opt_res.cif	The 3D structure obtained by fitting, in CIF file format.

Input parameters	Information
setting_abc	Lattice constants of the structure. [a,b,c,alpha,beta,gamma]

	Example: setting_abc=[5.5, 5.5, 8.5,90,90,90]
setting_maxeval	Number of fitting steps. Example: setting_maxeval=10000*3